

PHIL 450: Midterm

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Tip: the questions towards the end are harder, so do not spend longer than necessary on the earlier questions.

1. (a) Construct the following sequences using the two rules for constructing sequences of numbers in $\mathbb{N}^*$, translating them into a suitable notation if necessary. (5 points)
 - i. $(3 : \epsilon)$
 - ii. $(4, 5, 6)$
 - iii. $(3 : (2 : \epsilon))$
 - iv. (2)(b) Build three different trees containing exactly seven nodes, using the rules for constructing trees (nodes are include open and closed circles: $\circ$ and $\bullet$). (5 points)
2. State the rules for constructing sentences made from the connectives $\neg, \wedge$ and $\vee$ (i.e. the language $\mathcal{L}(\neg, \wedge, \vee)$).

Which of the following are sentences. If it is a sentence, construct it using the rules. (5 points).

- (a) $(\neg P)$
 - (b) $\neg(P \wedge Q)$
 - (c) $(\neg(P \wedge Q))$
 - (d) $(\neg(P \wedge Q) \vee (\neg P \vee R))$
3. (a) Define by recursion a function *Let* that maps a sentence of the propositional calculus to the number of propositional letters it contains. You may assume *add* has been defined. (3 points).
 - (b) Prove by induction that every sentence contains at least one propositional letter: $Let(A) > 0$ for every sentence A . (7 points.)
 - (c) Suppose that the propositional letters P, Q, R have meanings — e.g. $P =$ snow is white, $Q =$ London is in France etc. We will use the numbers 1 and 0 to stand for the possible truth values a sentence can have. The *truth value* of a sentence A is 1 if that sentence is true, and 0 if it is false.
Suppose v is a function mapping sentences to truth values.

- i. Find a formula that expresses the truth value of $(A \vee B)$ in terms of the truth value of A and the truth value of B . That is, fill in the equation: $v((A \vee B)) = \dots v(A) \dots v(B) \dots$.
Do the same for $(A \wedge B)$ and $\neg A$. $v((A \wedge B)) = \dots v(A) \dots v(B) \dots$
 $v(\neg A) = \dots v(A) \dots$. In this question you may help yourself to the binary operations $\min(n, m)$ and $\max(n, m)$ which take two numbers and give you the smallest and biggest respectively. You may also help yourself to the operation *flip* that switches 1 and 0. (5 points.)
 - ii. Suppose that, as it happens, all the propositional letters are true. Define by recursion a function v that maps arbitrary sentences of the propositional calculus to the truth value of those sentences. (5 points.)
4. In this question you may assume that the relation $m \leq n$ as already been defined: you may assume that you have defined whether one number less than or equal to another.
 - (a) Consider the operation, $\text{insert} : \mathbb{N} \times \mathbb{N}^* \rightarrow \mathbb{N}^*$ where $\text{insert}(m, t)$ is the result of inserting m into the sequence t after the last number in t that m is at least as big as. So $\text{insert}(4, (1, 3, 2, 8, 4, 5)) = (1, 3, 2, 4, 8, 4, 5)$: the 4 has been inserted after the 1, 3 and 2, because it is at least as big as 1, 3 and 2, and it has not been inserted any later, because 8 is bigger than it.

Using the informal description above, calculate $\text{insert}(3, (2, 3, 4))$ and $\text{insert}(5, (4, 1, 2, 3, 1, 11))$ (3 points)
 - (b) Define $\text{insert}(m, t)$ by recursion. Hint: you should hold m fixed, and define it by recursion on t . You will define it by cases, and need to appeal to the previous questions. (10 points).
 - (c) Define by recursion an operation, $\text{sort} : \mathbb{N}^* \rightarrow \mathbb{N}^*$, that sorts a sequence of numbers in ascending order. So, for instance, $\text{sort}((4, 1, 9, 8, 3, 4, 3, 1, 4)) = (1, 1, 3, 3, 4, 4, 4, 8, 9)$. After you have provided your answer, you should check that it works by calculating what it does to the sequence just given: $(4, 1, 9, 8, 3, 4, 3, 1, 4)$. You may assume that your definition of *insert* behaves as it ought to. (Hint: you will only need to use your definition of *insert* above.) (10 points).
5. In this question you may assume that multiplication $\text{mult} : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ has already been defined. (And you may use infix or prefix notation to represent it, as you prefer.)
 - (a) Using recursion, define the exponentiation function $\text{exp} : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ where $\text{exp}(m, n) = m^n = m \times m \times \dots \times m$, the result of multiplying m with itself n times. (5 points).

- (b) Consider the function f that maps the number n to the result of powering 2 to itself n times. We will simply stipulate that $f(0) = 1$. That is, $f(0) = 1$, $f(1) = 2$, $f(2) = 2^2 (= 4)$, $f(3) = 2^{2^2} (= 2^4 = 16)$, $f(4) = 2^{2^{2^2}}$ and so on. Now define it using recursion. (10 points).
- (c) Prove by induction that $\text{exp}(m, \text{add}(k, n)) = \text{mult}(\text{exp}(m, k), \text{exp}(m, n))$. (Hint: fix m and k and do induction on n .)
- You may assume any properties of multiplication and addition proved in exercises or in class. For instance, associativity of multiplication: $\text{mult}(\text{mult}(m, k), n) = \text{mult}(m, \text{mult}(k, n))$. (15 points.)